

Monitoring Tumorigenesis and Senescence In Vivo with a p16INK4a-Luciferase Model

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Evidence abstract

Monitoring cancer and aging in vivo remains experimentally challenging.

Here, we describe a luciferase knockin mouse (p16(LUC)), which faithfully reports expression of p16(INK4a), a tumor suppressor and aging biomarker.

Lifelong assessment of luminescence in p16(+LUC) mice revealed an exponential increase with aging, which was highly variable in a cohort of contemporaneously housed, syngeneic mice.

Expression of p16(INK4a) with aging did not predict cancer development, suggesting that the accumulation of senescent cells is not a principal determinant of cancer-related death.

In 14 of 14 tested tumor models, expression of p16(LUC) was focally activated by early neoplastic events, enabling visualization of tumors with sensitivity exceeding other imaging modalities.

Activation of p16(INK4a) was noted in the emerging neoplasm and surrounding stromal cells.

This work suggests that p16(INK4a) activation is a characteristic of all emerging cancers, making the p16(LUC) allele a sensitive, unbiased reporter of neoplastic transformation.

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Evidence checklist

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Source continuity

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